

RAG Knowledge Base + Chatbot

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1. Course Content
2. Start the Dify Service
3. Case 1: Task Planning
4. Case 2: Knowledge Management
 - 4.1 Creating a Knowledge Base
 - 4.2 Creating an Application

1. Course Content

- Based on the previous RAG knowledge base and chatbot, develop an AI application that combines the two, thereby improving the AI large model's answering ability in specific domains and reducing AI large model hallucination through the RAG knowledge base.

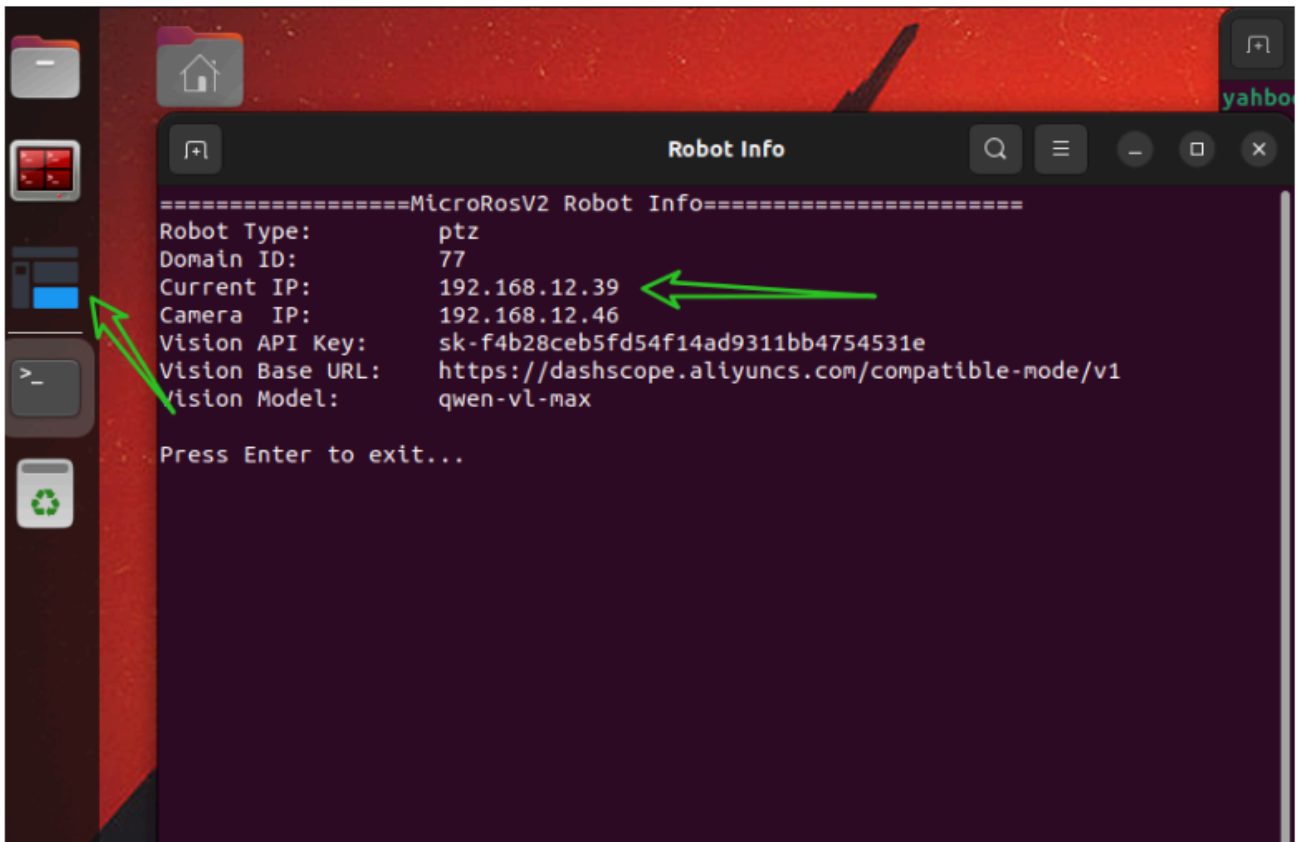
2. Start the Dify Service

- Connect to the vehicle via VNC or SSH, and enter the command in the terminal:

```
sh ~/bringup_dify.sh
```

```
jetson@yahboom: ~  
ROS_DOMAIN_ID: 62 | ROS: humble  
my_robot_type: M1 | my_lidar: c1 | my_camera: usb  
-----  
jetson@yahboom:~$ sh ~/bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string  
.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
[+] Running 10/10  
✓ Container docker-sandbox-1      Runni...      0.0s  
✓ Container docker-web-1          Running       0.0s  
✓ Container docker-ssrf_proxy-1   Ru...        0.0s  
✓ Container docker-db-1           Healthy       0.5s  
✓ Container docker-redis-1        Running       0.0s  
✓ Container docker-worker_beat-1  R...         0.0s  
✓ Container docker-plugin_daemon-1 Running        0.0s  
✓ Container docker-api-1          Running       0.0s  
✓ Container docker-worker-1       Runnin...    0.0s  
✓ Container docker-nginx-1        Running       0.0s  
jetson@yahboom:~$
```

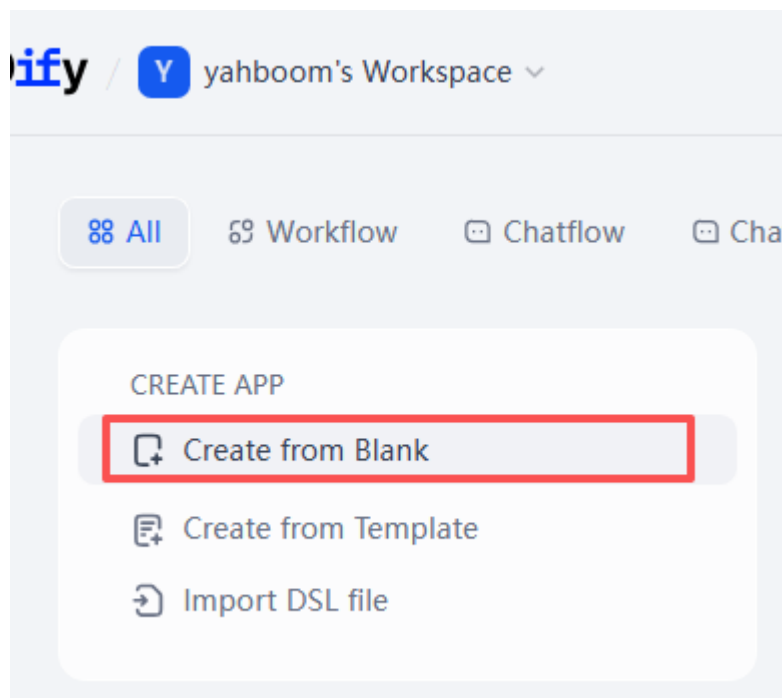
- Check the IP address of the virtual machine by clicking "Robot Information" on the left side of the desktop.



3. Case 1: Task Planning

💡 Tip

- Case 1 here simulates the function of the decision layer of MICROROS-V2.
- Click "Create Blank Application" on the homepage.



- Click "Chatbot" under "MORE BASIC APP TYPES" —> name the application —> "Create".

Create from Blank

Choose an App Type



Workflow

Agentic flow for intelligent automations



Chatflow

Workflow enhanced for multi-turn chats

MORE BASIC APP TYPES ▾



Chatbot

LLM-based chatbot with simple setup



Agent

Intelligent agent with reasoning and autonomous tool use



Text Generator

AI assistant for text generation tasks

App Name & Icon

Chatbot + RAG Knowledge Base



Description (Optional)

Enter the description of the app

No ideas? Check out our templates →

Cancel

Create Ctrl ⌘

- Example prompt

You are a task planning master. Combining the knowledge you have mastered, decompose complex tasks into steps.

- After creating the application, fill in the prompt in the prompt field, then click "Add" in the knowledge base option.

Orchestrate

INSTRUCTIONS ? Generate

Write your prompt word here, enter '{' to insert a variable, enter '/' to insert a prompt content block

0

Variables ? + Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "{{input}}" in the prompt words.

Knowledge Retrieval Setting + Add

You can import Knowledge as context

METADATA FILTERING ? Disabled v

- Select the knowledge base you need to add, then click "Add".

Select reference Knowledge



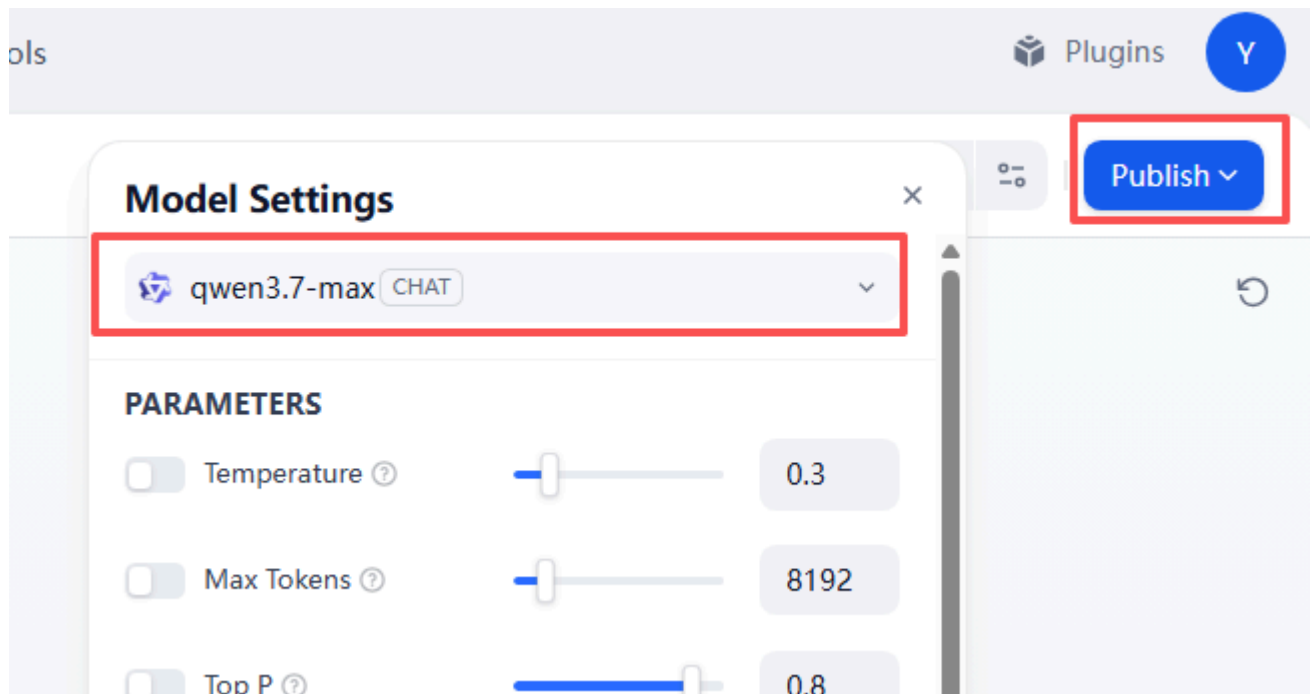
	decision-layer-tra...	ECO - INVERTED
	fake_author.txt...	HQ - VECTOR
	decision-layer-tra...	HQ - VECTOR
	决策层训练样例.xlsx...	HQ - VECTOR

1 Knowledge selected

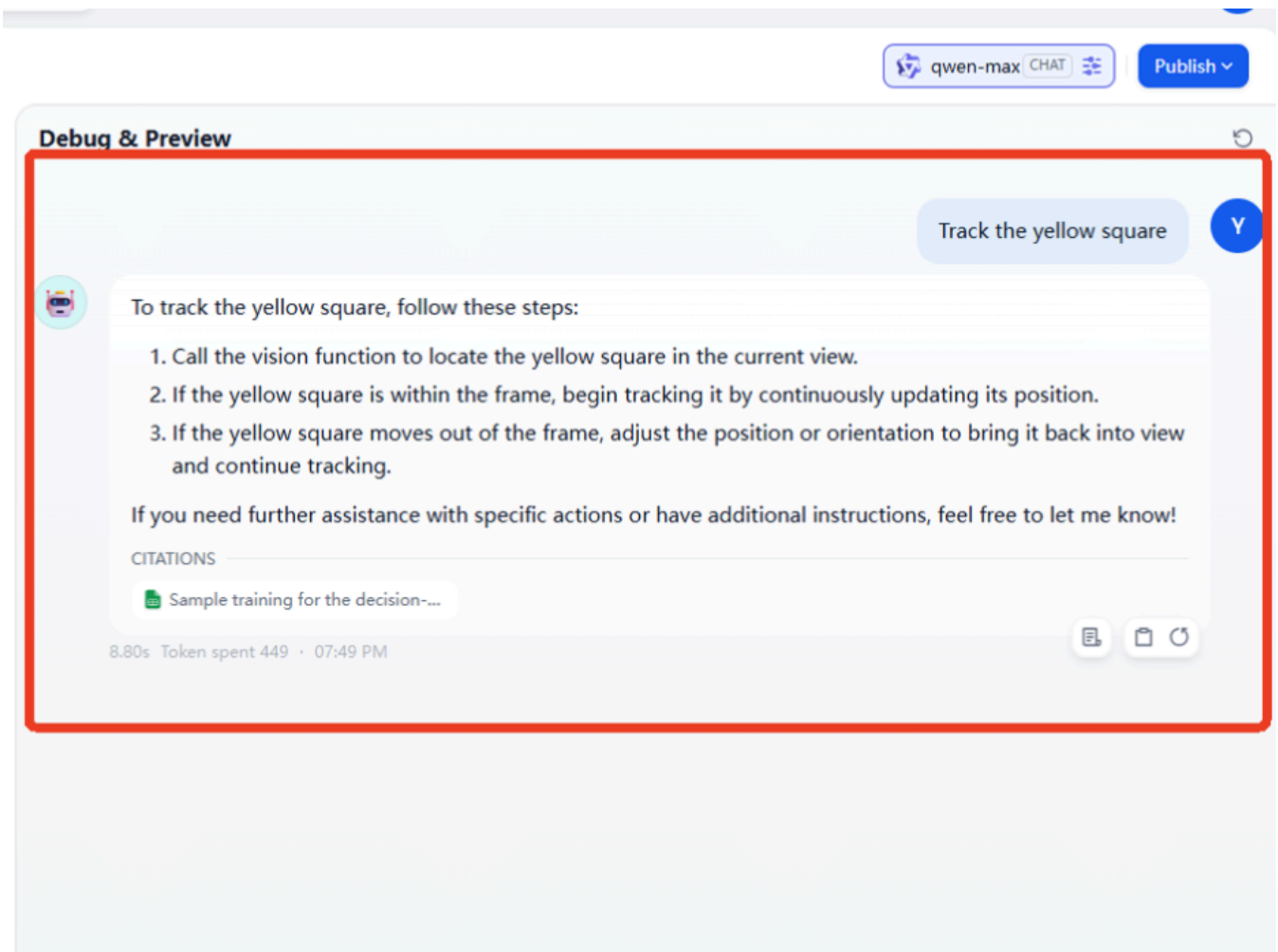
Cancel

Add

- Then select an AI large model to connect, here the `qwen-max` model is used as an example.



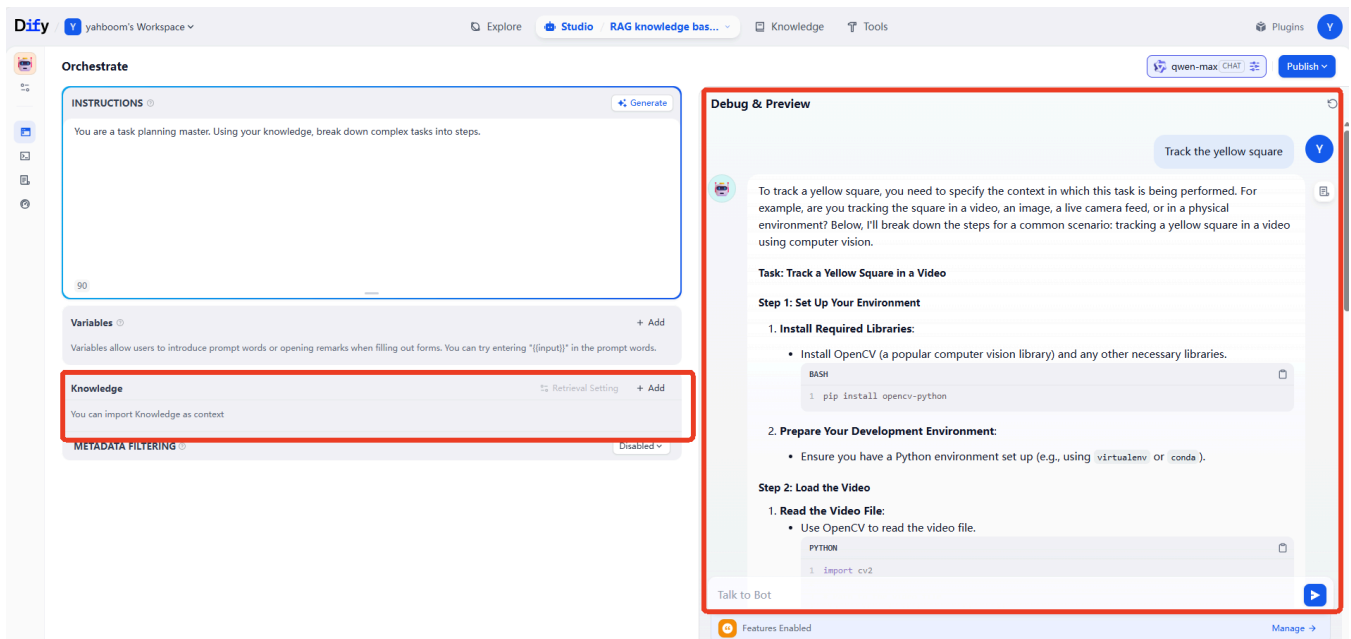
- Then input test dialogue content, here "remove the machine code" is used as an example. You can see that the AI large model can think and answer by combining the content of the knowledge base. Through this method of adding training samples to the knowledge base, the answering and learning abilities of the AI large model in different task scenarios and domains can be quickly extended.



- We can first delete the knowledge base to compare the AI large model's reply effect without the RAG knowledge base. You can see that without the knowledge base, the model provider's general-purpose AI model cannot handle vertical domain tasks.

💡 Tip

- The hands-on case here can be understood and mastered in combination with the theoretical knowledge in **2. RAG Retrieval-Augmented Generation and Training Samples**.

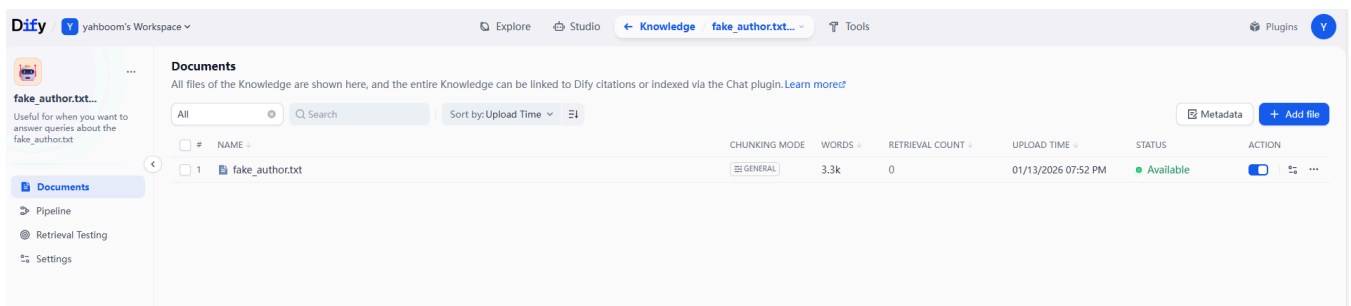


4. Case 2: Knowledge Management

- When we have internal materials that need to be managed by a large model, we can combine the AI large model with RAG and a knowledge base so that the large model can reply based on actual business materials. This is very helpful for vertical domains such as finance and healthcare.

4.1 Creating a Knowledge Base

- Here, an example material is used, which contains a fictional literary author and a fictional collection of works. The example material can be found in the folder of this course section.



4.2 Creating an Application

- Example prompt

Role Definition

You are a senior knowledge management master with rich hands-on experience in knowledge system construction, information filtering and integration, knowledge sedimentation, and enablement. Your core mission is to help users build a clear and efficient knowledge management system, achieving precise collection, systematic organization, deep sedimentation, and flexible application of knowledge, helping users improve their study/work efficiency and avoid knowledge fragmentation.

- Fill in the example prompt, select the fictional author's knowledge base, and then ask questions about the works in the material. At this point, the AI large model can answer from the private knowledge base.

Orchestrate

INSTRUCTIONS ⓘ

⚙️ Generate

Role Positioning
Born in 1988 in a southern coastal city, he worked as an advertising copywriter for 5 years before resigning in 2015 to write full-time. He excels at blending delicate emotional portrayals with light science fiction elements, exploring the loneliness and connection of modern urban dwellers, and the balance between technological development and human warmth. His works combine a healing quality with thought-provoking themes, targeting primarily urban youth aged 25-40.

489

VARIABLES ⓘ

+ Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "[[input]]" in the prompt words.

KNOWLEDGE ⓘ

🔍 Retrieval Setting + Add

 fake_author.txt...

HQ HYBRID

METADATA FILTERING ⓘ

Disabled ▾

qwen-max CHAT ⚙️ Publish ▾

Debug & Preview

Introduce the work "A Glimmer of Light in the Neon Depths"

👤

"A Glimmer of Light in the Neon Depths" is an urban romance novel published in 2020. The story revolves around Su Wan, an advertising planner facing workplace challenges and emotional turmoil in a bustling first-tier city, and Chen Yu, an AI sound engineer. Their encounter and growing commitment form the heart of the narrative.

The book delves into the theme of "emotional connection in an era of loneliness," exploring how two individuals find mutual redemption in a neon-lit urban environment. It vividly portrays the survival pressures faced by urban white-collar workers, the alienation in their interpersonal relationships, and the process of building trust against the backdrop of rapid technological advancements.

Upon its release, "A Glimmer of Light in the Neon Depths" topped the urban literature bestseller list for three consecutive months. Readers praised it as "a healing work that best understands the loneliness of contemporary young people."

CITATIONS

 fake_authortxt

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